SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	Narappagari Sravya	**Reg. No.:**	192472192
Section / Batch:	B.E CSE-AI / 2024	**Date:**	31-07-2026

Code of Conduct

I **Narappagari Sravya(192472192)***(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:sravya

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words “talked”, “talking”, and “talks”. Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.

2. Given the input words: “relational”, “conditional”, “happiness”, “running”, design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.

3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as “optimize”, “optimization”, “optimized”, and “optimizing”. The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.

4. A large-scale information retrieval system processes input words: “compute”, “computer”, “computing”, and “computation” to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.

5. A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Inflectional Morphology & Porter Stemmer – talked, talking, talks

Aim

To apply inflectional morphology and Porter Stemmer rules to obtain the normalized root form.

Input Word	Suffix Removed	Type	Porter Stemmer	Normalized Root
talked	-ed	Inflectional	talked → talk	talk
talking	-ing	Inflectional	talking → talk	talk
talks	-s	Inflectional	talks → talk	talk

Role of Morphological Normalization

- Groups different grammatical forms into one base word.
- Reduces duplicate index entries.
- Improves search accuracy and document retrieval.
- Makes keyword matching more efficient.

Result

All words are normalized to **talk**.

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2. Hybrid Morphological Pipeline – relational, conditional, happiness, running

Aim

To combine derivational analysis, inflectional analysis, finite-state parsing, and Porter Stemming for root extraction.

Input Word	Root	Affix	Type	Finite-State Transition	Final Stem
relational	relate	-ional	Derivational	Start → relate → relational	relat
conditional	condition	-al	Derivational	Start → condition → conditional	condit
happiness	happy	-ness	Derivational	Start → happy → happiness	happi
running	run	-ing	Inflectional	Start → run → running	run

Technique Contribution

- **Derivational Rules:** Identify word formation.
- **Inflectional Analysis:** Remove grammatical endings.
- **Finite-State Parsing:** Validates word transformations.
- **Porter Stemmer:** Produces the final normalized stem.

Result

Final stems: **relat, condit, happi, run**

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3. SEO Morphological Processing – optimize, optimization, optimized, optimizing

Aim

To normalize keyword variations for consistent indexing and search ranking.

Input Word	Root	Affix	Type	Finite-State Parsing	Final Stem
optimize	optimize	—	Base Word	Start → optimize	optim
optimization	optimize	-ation	Derivational	optimize → optimization	optim
optimized	optimize	-ed	Inflectional	optimize → optimized	optim
optimizing	optimize	-ing	Inflectional	optimize → optimizing	optim

Benefits

- Removes duplicate keyword entries.
- Improves search relevance.

- Enhances indexing efficiency.
- Maintains semantic consistency.

Result

All words are normalized to the stem **optim**.

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4. Hybrid Morphological Pipeline – compute, computer, computing, computation

Aim

To normalize related words using morphological analysis, finite-state parsing, and Porter Stemming.

Input Word	Root	Affix	Type	State Transition	Final Stem
compute	compute	—	Base Word	Start → compute	comput
computer	compute	-er	Derivational	compute → computer	comput
computing	compute	-ing	Inflectional	compute → computing	comput
computation	compute	-ation	Derivational	compute → computation	comput

Result

All words are normalized to the common stem **comput** for efficient indexing.

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5. Morphology-Based Framework – decide, decision, decisive, deciding

Aim

To combine derivational rules, inflectional transformations, finite-state parsing, and Porter Stemming.

Input Word	Root	Affix	Type	State Transition	Final Stem
decide	decide	—	Base Word	Start → decide	decid
decision	decide	-sion	Derivational	decide → decision	decis
decisive	decide	-ive	Derivational	decide → decisive	decis
deciding	decide	-ing	Inflectional	decide → deciding	decid

Effect of Suffixes

- **-sion** forms a noun (**decision**).
- **-ive** forms an adjective (**decisive**).

- **-ing** forms the present participle (**deciding**).
- The base meaning (**decide**) remains the same.

Result

The normalized stems are:

- **decide → decid**
- **decision → decis**
- **decisive → decis**
- **deciding → decid**